

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

(Ancillary Subject)

Paper: BCS-203 — Fundamentals of Computers

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions including Question 1, which is compulsory.***Question 1. (Compulsory)**

- (a) Why do we use secondary memory? [3.5]
- (b) What are the two main components of CPU in a computer system? List the main functions of each component. [3.5]
- (c) Why are peripheral devices necessary for a computer system? List any five peripheral devices. [3.5]
- (d) Mention the different types of software used in computer systems. Explain each with an example. [3.5]

Question 2.

- (a) Draw the block diagram of basic organization of a computer system. Explain each component with an example. [7]
- (b) What is primary storage device? Explain RAM and ROM. [5]
- (c) How many number of bytes are in a Kilobyte, Megabyte, Gigabyte and Terabyte? [2]

Question 3.

- (a) Write short notes on the Generations of computers. [6]
- (b) Convert the following numbers according to the specified bases: [8]
 - (i) $(11010011)_2 = (?)_{16}$
 - (ii) $(111011.10)_2 = (?)_{10}$
 - (iii) $(A6B4)_{16} = (?)_8$
 - (iv) $(123)_8 = (?)_{10}$

Question 4.

- (a) Explain the applications of computer in various fields. [6]
- (b) What is an operating system? What are the main functions of an operating system? [8]

Question 5.

- (a) What are the major components of database management systems? Describe each of those components. [8]
- (b) What are input devices? Explain any three input devices. [6]

Question 6.

- (a) Answer the following multiple choice questions: [2×4 = 8]
- (i) Which access method is used for magnetic tape?
(i) Direct (ii) Random (iii) Sequential (iv) None of the above
- (ii) The language that the computer can understand and execute is called
(i) Machine Language (ii) Application software (iii) System program (iv) None of the above
- (iii) Which of the following is used as a primary storage device?
(i) Magnetic tape (ii) PROM (iii) Floppy disk (iv) None of the above
- (iv) Operating system is
(i) A collection of hardware components (ii) A collection of software routines
(iii) A collection of Input-Output device (iv) None of the above
- (b) Write the differences between serial access memory and random access memory. List any two secondary storage devices with their characteristics. [6]

Question 7. Write short notes on any *two* of the following:

[7×2 = 14]

- (a) Instruction Cycle
- (b) Cache Memory
- (c) BCD code